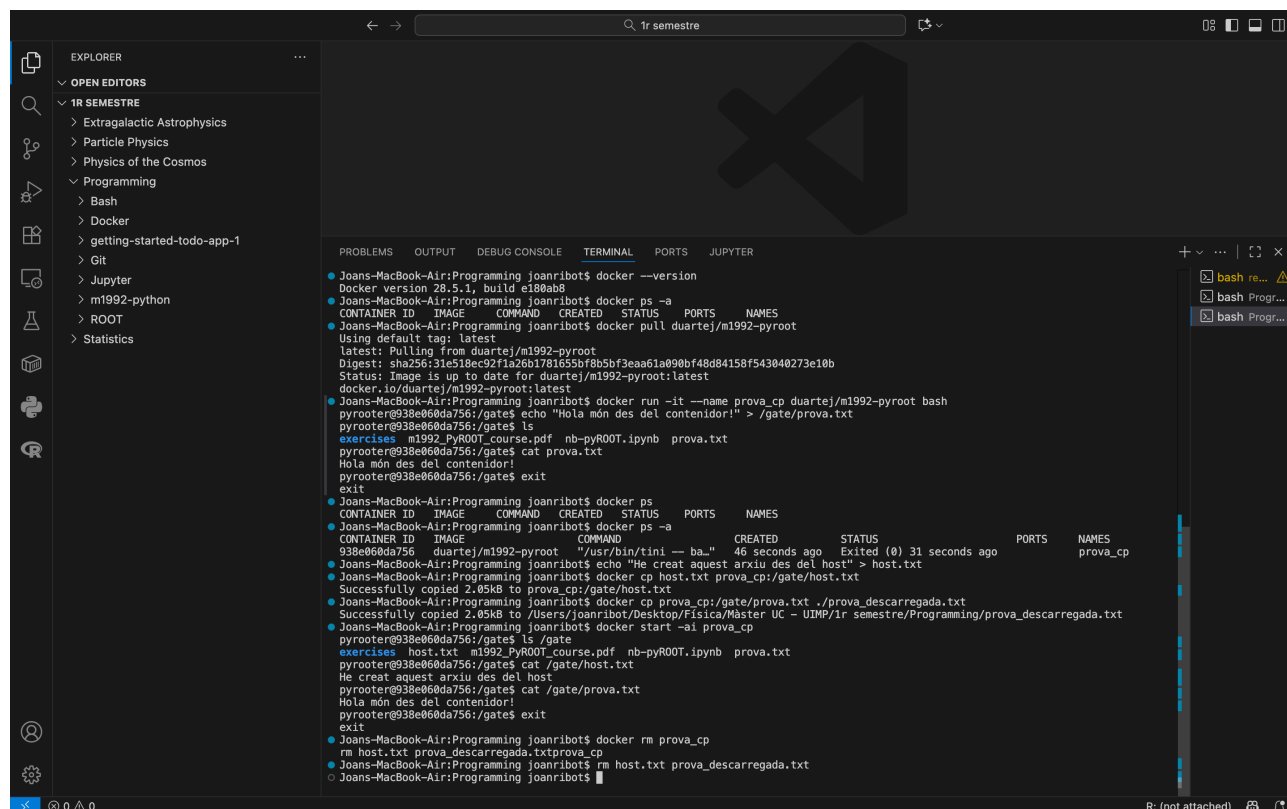


Exercise 1.

Share data between the host and containers

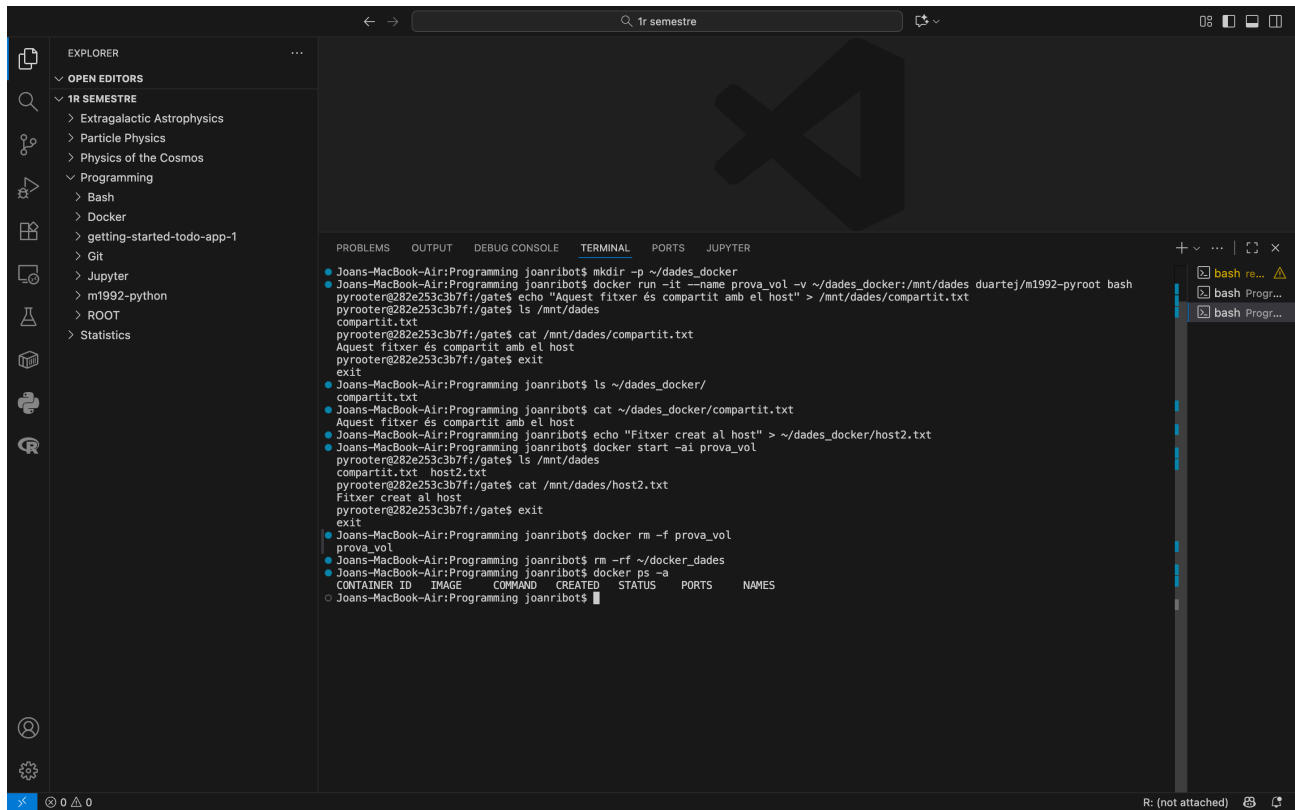
Para empezar compruebo que docker está instalado y funciona correctamente.



```
Joans-MacBook-Air:Programming joanribot$ docker --version
Docker version 28.5.1, build e188ab8
Joans-MacBook-Air:Programming joanribot$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS        NAMES
Joans-MacBook-Air:Programming joanribot$ docker pull duartej/m1992-pyroot
latest: Pulling from duartej/m1992-pyroot
Digest: sha256:31e518ec92f1a26b1781655bf8b5bf3eaa61a090bf48d84158f543040273e10b
Status: Image is up to date for duartej/m1992-pyroot:latest
Using default tag: latest
Joans-MacBook-Air:Programming joanribot$ docker run -it --name prova_cp duartej/m1992-pyroot bash
pyroot@938e060da756:/gate$ echo "Hola món des del contenidor!" > /gate/prova.txt
pyroot@938e060da756:/gate$ ls
exercises  m1992_PyROOT_course.pdf  nb-pyROOT.ipynb  prova.txt
pyroot@938e060da756:/gate$ cat prova.txt
Hola món des del contenidor!
pyroot@938e060da756:/gate$ exit
exit
Joans-MacBook-Air:Programming joanribot$ docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS        NAMES
Joans-MacBook-Air:Programming joanribot$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS        NAMES
938e060da756   duartej/m1992-pyroot  "/usr/bin/tini -- ba..."  46 seconds ago  Exited (0) 31 seconds ago        prova_cp
Joans-MacBook-Air:Programming joanribot$ docker cp host.txt prova_cp:/gate/host.txt
Successfully copied 2.05kB to prova_cp:/gate/host.txt
Joans-MacBook-Air:Programming joanribot$ docker cp prova_cp:/gate/prova.txt ./prova_descarregada.txt
Successfully copied 2.05kB to /Users/joanribot/Desktop/Física/Master UC - UIMP/1r semestre/Programming/prova_descarregada.txt
Joans-MacBook-Air:Programming joanribot$ docker start -ai prova_cp
pyroot@938e060da756:/gate$ ls /gate
exercises  host.txt  m1992_PyROOT_course.pdf  nb-pyROOT.ipynb  prova.txt
pyroot@938e060da756:/gate$ cat /gate/host.txt
He creat aquest arxiu des del host
pyroot@938e060da756:/gate$ cat /gate/prova.txt
Hola món des del contenidor!
pyroot@938e060da756:/gate$ exit
exit
Joans-MacBook-Air:Programming joanribot$ docker rm prova_cp
rm host.txt prova_descarregada.txtprova_cp
Joans-MacBook-Air:Programming joanribot$ rm host.txt prova_descarregada.txt
Joans-MacBook-Air:Programming joanribot$
```

Creo el contenedor a partir de la imagen, genero un archivo .txt de prueba dentro del contenedor para posteriormente copiarlo en mi disco local. Posteriormente realizo el proceso inverso: creo un archivo en el disco local y lo copio en el contenedor, que debe estar ejecutándose, con el comando 'cp' y el 'path' o camino correspondiente.

La otra opción es enlazar el contenedor al host local al ejecutarlo.



The screenshot shows the Visual Studio Code interface with a terminal window open. The Explorer sidebar on the left shows a project structure under '1R SEMESTRE' including folders like 'Extragalactic Astrophysics', 'Particle Physics', 'Physics of the Cosmos', 'Programming', 'Bash', 'Docker', and files like 'getting-started-todo-app-1', 'Git', 'Jupyter', 'm1992-python', 'ROOT', and 'Statistics'.

The terminal window displays the following commands and output:

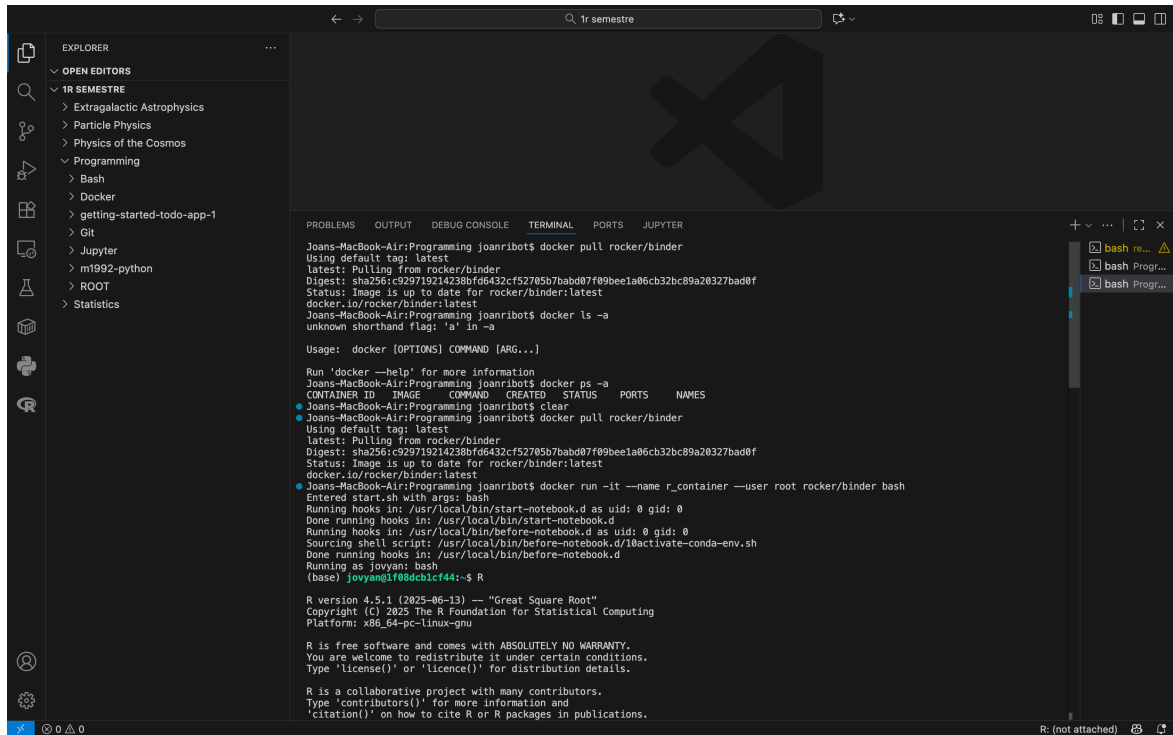
```
Joans-MacBook-Air:Programming joanribot$ mkdir -p ~/dades_docker
Joans-MacBook-Air:Programming joanribot$ docker run --name prova_vol -v ~/dades_docker:/mnt/dades duartej/m1992-pyroot bash
pyrooter@282e253c3b7f:/gate$ echo "Aquest fitxer és compartit amb el host" > /mnt/dades/compartit.txt
pyrooter@282e253c3b7f:/gate$ ls /mnt/dades
compartit.txt
pyrooter@282e253c3b7f:/gate$ cat /mnt/dades/compartit.txt
Aquest fitxer és compartit amb el host
pyrooter@282e253c3b7f:/gate$ exit
exit
Joans-MacBook-Air:Programming joanribot$ ls ~/dades_docker/
compartit.txt
Joans-MacBook-Air:Programming joanribot$ cat ~/dades_docker/compartit.txt
Aquest fitxer és compartit amb el host
Joans-MacBook-Air:Programming joanribot$ echo "Fitxer creat al host" > ~/dades_docker/host2.txt
Joans-MacBook-Air:Programming joanribot$ docker start -ai prova_vol
pyrooter@282e253c3b7f:/gate$ ls /mnt/dades
compartit.txt  host2.txt
pyrooter@282e253c3b7f:/gate$ cat /mnt/dades/host2.txt
Fitxer creat al host
pyrooter@282e253c3b7f:/gate$ exit
exit
Joans-MacBook-Air:Programming joanribot$ docker rm -f prova_vol
prova_vol
Joans-MacBook-Air:Programming joanribot$ rm -rf ~/dades_dades
Joans-MacBook-Air:Programming joanribot$ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
Joans-MacBook-Air:Programming joanribot\$						

Exercise 2.

Installing missing packages

Para empezar compruebo que la imagen de rocker/binder está descargada. A continuación ejecuto un contenedor (sin -rm para no eliminarlo) y entro en R, donde instalo los paquetes Hmisc (dentro del contenedor)



```
Joans-MacBook-Air:Programming joanribots docker pull rocker/binder
Using default tag: latest
latest: Pulling from rocker/binder
Digest: sha256:c929719214238bfd6432cf52705b7babb07f09beela06cb32bc89a26327bad0f
Status: Image is up to date for rocker/binder:latest
Joans-MacBook-Air:Programming joanribots docker ls -a
unknown shorthand flag: 'a' in -a

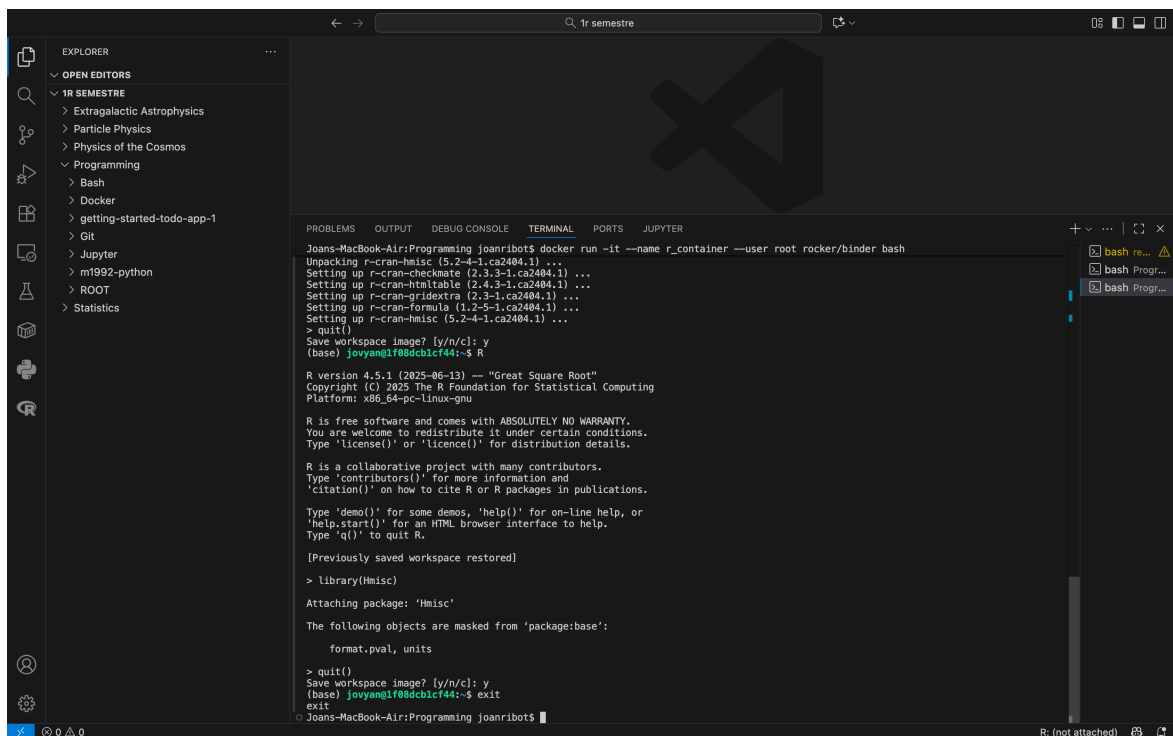
Usage: docker [OPTIONS] COMMAND [ARG...]

Run 'docker --help' for more information
Joans-MacBook-Air:Programming joanribots docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
Joans-MacBook-Air:Programming joanribots clear
Joans-MacBook-Air:Programming joanribots docker pull rocker/binder
Using default tag: latest
latest: Pulling from rocker/binder
Digest: sha256:c929719214238bfd6432cf52705b7babb07f09beela06cb32bc89a26327bad0f
Status: Image is up to date for rocker/binder:latest
Joans-MacBook-Air:Programming joanribots docker run -it --name r_container --user root rocker/binder bash
Entered start.sh with args: bash
Running hooks in: /usr/local/bin/start-notebook.d as uid: 0 gid: 0
Done running hooks in: /usr/local/bin/start-notebook.d
Running hooks in: /usr/local/bin/before-notebook.d as uid: 0 gid: 0
Sourcing shell script: /usr/local/bin/before-notebook.d/10activate-conda-env.sh
Done running hooks in: /usr/local/bin/before-notebook.d
Running as jovyana: bash
(base) jovyana@f08dcb1cf44:~$ R

R version 4.5.1 (2025-06-13) -- "Great Square Root"
Copyright (C) 2025 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.
```



```
Joans-MacBook-Air:Programming joanribots docker run -it --name r_container --user root rocker/binder bash
Unpacking r-cran-hmisc (5.2-4-1.ca2404.1) ...
Setting up r-cran-checkmate (2.3.3-1.ca2404.1) ...
Setting up r-cran-hintable (2.4.3-1.ca2404.1) ...
Setting up r-cran-gridextra (2.3-1.ca2404.1) ...
Setting up r-cran-formula (1.2-5-1.ca2404.1) ...
Setting up r-cran-hmisc (5.2-4-1.ca2404.1) ...
> quit()
Save workspace image? [y/n/c]: y
(base) jovyana@f08dcb1cf44:~$ R

R version 4.5.1 (2025-06-13) -- "Great Square Root"
Copyright (C) 2025 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Previously saved workspace restored]
> library(Hmisc)

Attaching package: 'Hmisc'

The following objects are masked from 'package:base':

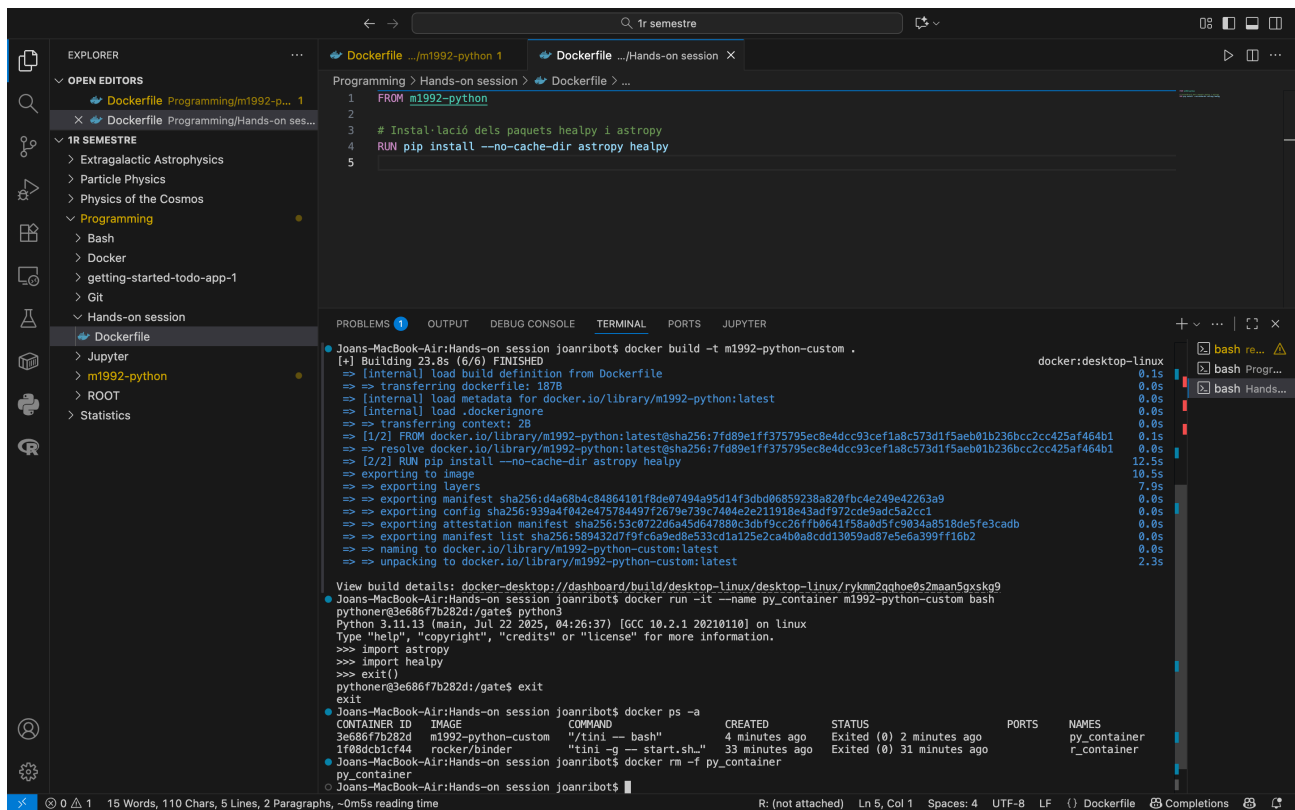
    format.pval, units

> quit()
Save workspace image? [y/n/c]: y
(base) jovyana@f08dcb1cf44:~$ exit
exit
Joans-MacBook-Air:Programming joanribots
```

Finalmente compruebo que los paquetes funcionan y salgo del contenedor. Mientras este no se elimine y esté activo (en segundo plano), los cambios no se pierden. En caso contrario todo se

elimina y se debe hacer la instalación de nuevo.

A partir de la imagen 'm1992-python' creo un contenedor y instalo los paquetes de 'astropy' y 'healpy' permanentemente modificando el Dockerfile. De esta manera, cada vez que se ejecute un contenedor a partir de esta imagen los paquetes ya estarán instalados.



The screenshot shows a VS Code editor with a Dockerfile open in the editor pane and its build output in the terminal pane. The Dockerfile is located at `.../m1992-python 1` and contains the following content:

```
1 FROM m1992-python
2
3 # Instal·lació dels paquets healpy i astropy
4 RUN pip install --no-cache-dir astropy healpy
5
```

The terminal pane shows the output of the `docker build` command, which successfully builds the image `m1992-python-custom`. The output includes the following information:

- Building 23.8s (6/6) FINISHED
- [+] Building 23.8s (6/6) FINISHED
- [internal] load build definition from Dockerfile
- => transferring dockerfile: 187B
- [internal] load metadata for docker.io/library/m1992-python:latest
- [internal] load .dockerignore
- => transferring context: 2B
- [1/2] FROM docker.io/library/m1992-python:latest@sha256:7fd89e1ff375795ec8e4d9c93cef1a8c573d1f5aeb01b236bcc2cc425af464b1
- => resolve docker.io/library/m1992-python:latest@sha256:7fd89e1ff375795ec8e4d9c93cef1a8c573d1f5aeb01b236bcc2cc425af464b1
- [2/2] RUN pip install --no-cache-dir astropy healpy
- => exporting to image
- => exporting layers
- => exporting manifest sha256:d4a68b4c84864101f8de87494a95d14f3dbd06859238a820fbc4e249e42263a9
- => exporting config sha256:939a4f042e475784497f2679e739c7404e2e211918e43adf972cde9adc5a2cc1
- => exporting attestation manifest sha256:53c0722d6a45d647880c3dbf9cc26fffb0641f58a0d5fc9834a8518de5fe3cadb
- => exporting manifest list sha256:58943207f9fc6a9ed8e533cd1a125e2ca4b8a8cdd13859ad87e5ea399ff16b2
- => naming to docker.io/library/m1992-python-custom:latest
- => unpacking to docker.io/library/m1992-python-custom:latest

The terminal also shows the output of the `docker run` command, which successfully runs the container `py_container`. The output includes the following information:

- python3.11.13 (main, Jul 22 2025, 04:26:37) [GCC 10.2.1 20210110] on linux
- Type "help", "copyright", "credits" or "license()" for more information.
- >>> import astropy
- >>> import healpy
- >>> exit()
- python3.11.13 (main, Jul 22 2025, 04:26:37) [GCC 10.2.1 20210110] on linux
- Type "help", "copyright", "credits" or "license()" for more information.
- >>> import astropy
- >>> import healpy
- >>> exit()

The terminal also shows the output of the `docker ps` command, which lists the container `py_container` with the following information:

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
3e686f7b282d	m1992-python-custom	"/tini -- bash"	4 minutes ago	Exited (0) 2 minutes ago		py_container
1f08dc1cf44	rocker/binder	"tini -g -- start.sh."	33 minutes ago	Exited (0) 31 minutes ago		r_container

Exercise 3.

Detach mode

Al ejecutar un contenedor con el flag '-d' este corre directamente en segundo plano. Para adherirme uso 'attach' y para salir del contenedor sin eliminarlo uso la secuencia 'Ctrl+P' + 'Ctrl+Q'.

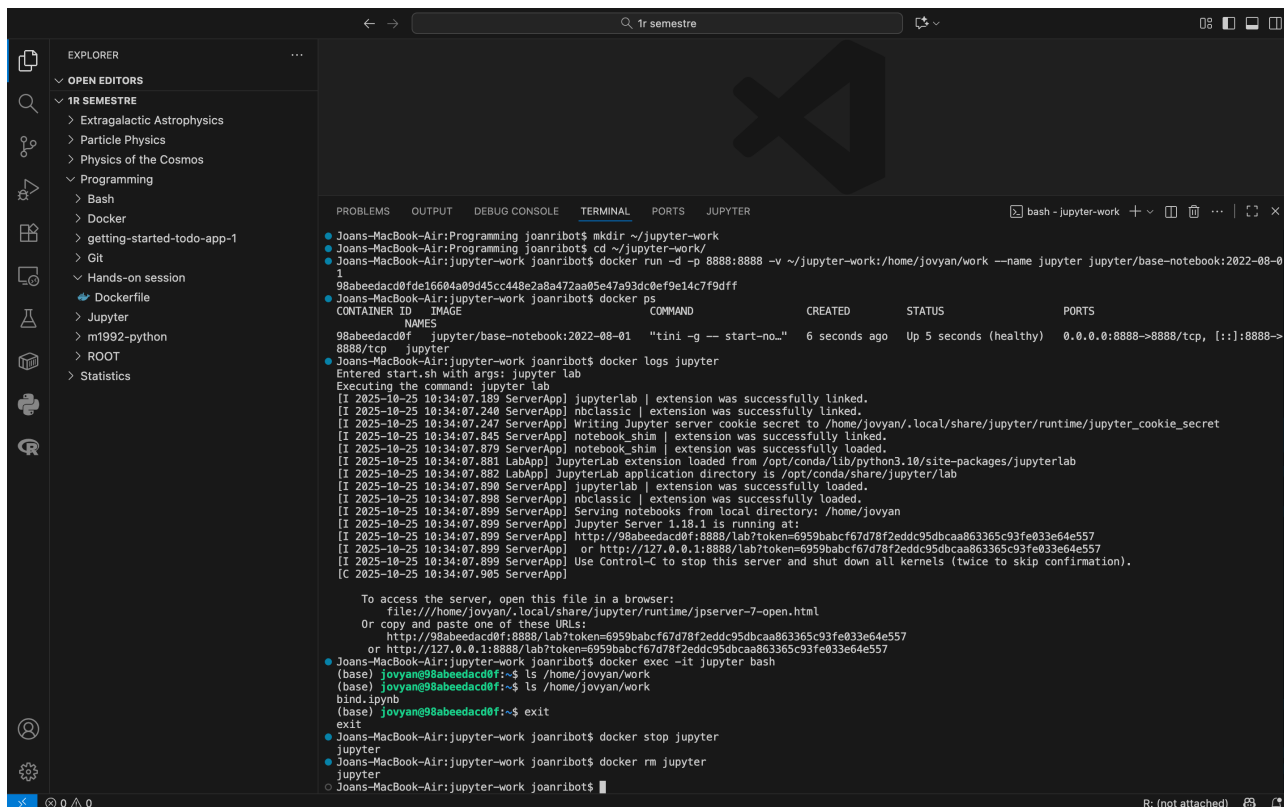
```
...ca/Máster UC - UIMP/tr semestre/Programming/Hands-on session -- pythoner@49c33b2561d4:/gate -- -bash
Joans-MacBook-Air:Hands-on session joanribot$ docker run -it --name int m1992-python
Python 3.11.13 (main, Jul 22 2025, 04:26:37) [GCC 10.2.1 20210110] on linux
Type "help", "copyright", "credits" or "license()" for more information.
>>>
>>> Per sortir d'aquí es pot fer exit() o Ctrl+P i Ctrl+Q
Joans-MacBook-Air:Hands-on session joanribot$ docker run -dit --name d-int m1992-python
d591a977f5f51f4379b652b18d6af3ad7db369372b55bb150f6e1a3816a1d812
Joans-MacBook-Air:Hands-on session joanribot$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
d591a977f5f51  m1992-python  "/tini -- python3"      4 seconds ago  Up 3 seconds  0.0.0.0:80->80  d-int
181ae9e6087d   m1992-python  "/tini -- python3"      About a minute ago  Up About a minute  0.0.0.0:80->80  int
Joans-MacBook-Air:Hands-on session joanribot$ docker attach d-int
>>> read escape sequence
Joans-MacBook-Air:Hands-on session joanribot$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
d591a977f5f51  m1992-python  "/tini -- python3"      39 seconds ago  Up 38 seconds  0.0.0.0:80->80  d-int
181ae9e6087d   m1992-python  "/tini -- python3"      2 minutes ago   Up 2 minutes   0.0.0.0:80->80  int
Joans-MacBook-Air:Hands-on session joanribot$ docker ps -q
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
d591a977f5f51  m1992-python  "/tini -- python3"      40 seconds ago  Up 40 seconds  0.0.0.0:80->80  d-int
181ae9e6087d   m1992-python  "/tini -- python3"      2 minutes ago   Up 2 minutes   0.0.0.0:80->80  int
Joans-MacBook-Air:Hands-on session joanribot$ docker stop int d-int
int
d-int
Joans-MacBook-Air:Hands-on session joanribot$ docker rm int d-int
int
d-int
Joans-MacBook-Air:Hands-on session joanribot$
```

En este caso he tenido que usar la terminal de macOS directamente en lugar de la incorporada en VS Code debido al solapamiento del comando 'Ctrl+Q'.

Exercise 4.

Create a jupyter notebook

Ya estoy de vuelta en la terminal de VS Code. Primero creo una carpeta para poder enlazar el disco local con el container con el 'jupyter notebook'. Luego ejecuto el contenedor, lo abro en el servidor local y genero bind.ipynb, un archivo de prueba. Después compruebo que no está porque lo he creado fuera de la carpeta /work, así que lo muevo dentro y compruebo que sí está. Finalmente salgo y elimino el contenedor.



The screenshot shows the VS Code interface with a terminal window open. The Explorer sidebar on the left shows a project structure with folders like '1R SEMESTRE', 'Programming', 'Bash', 'Docker', 'getting-started-todo-app-1', 'Git', 'Hands-on session', 'Dockerfile', 'Jupyter', 'm1992-python', 'ROOT', and 'Statistics'. The terminal window displays the following commands and output:

```
Joans-MacBook-Air:Programming joanribot$ mkdir ~/jupyter-work
Joans-MacBook-Air:Programming joanribot$ cd ~/jupyter-work/
Joans-MacBook-Air:jupyter-work joanribot$ docker run -d -p 8888:8888 -v ~/jupyter-work:/home/jovyan/work --name jupyter jupyter/base-notebook:2022-08-01
98abeeadacd0fde16604a09d45cc448e2a8a472aa05e47a93dc0ef9e14c7f9dfff
Joans-MacBook-Air:jupyter-work joanribot$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
98abeeadacd0f   jupyter/base-notebook:2022-08-01   "tini -g -- start-no..." 6 seconds ago  Up 5 seconds (healthy)  0.0.0.0:8888->8888/tcp, [::]:8888->8888/tcp
Joans-MacBook-Air:jupyter-work joanribot$ docker logs jupyter
Executing the command: jupyter lab
[I 2025-10-25 10:34:07.189 ServerApp] jupyterlab | extension was successfully linked.
[I 2025-10-25 10:34:07.240 ServerApp] nbclassic | extension was successfully linked.
[I 2025-10-25 10:34:07.247 ServerApp] Writing Jupyter server cookie secret to /home/jovyan/.local/share/jupyter/runtime/jupyter_cookie_secret
[I 2025-10-25 10:34:07.845 ServerApp] notebook_shim | extension was successfully linked.
[I 2025-10-25 10:34:07.879 ServerApp] notebook_shim | extension was successfully loaded.
[I 2025-10-25 10:34:07.881 LabApp] JupyterLab extension loaded from /opt/conda/lib/python3.10/site-packages/jupyterlab
[I 2025-10-25 10:34:07.882 LabApp] JupyterLab application directory is /opt/conda/share/jupyter/lab
[I 2025-10-25 10:34:07.898 ServerApp] jupyterlab | extension was successfully loaded.
[I 2025-10-25 10:34:07.898 ServerApp] nbclassic | extension was successfully loaded.
[I 2025-10-25 10:34:07.899 ServerApp] Serving notebooks from local directory: /home/jovyan
[I 2025-10-25 10:34:07.899 ServerApp] Jupyter Server 1.10.1 is running at:
[I 2025-10-25 10:34:07.899 ServerApp] http://98abeeadacd0f:8888/lab?token=6959babcf67d78f2eddc95dbcaa863365c93fe033e64e557
[I 2025-10-25 10:34:07.899 ServerApp] or http://127.0.0.1:8888/lab?token=6959babcf67d78f2eddc95dbcaa863365c93fe033e64e557
[I 2025-10-25 10:34:07.899 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
[C 2025-10-25 10:34:07.905 ServerApp]

To access the server, open this file in a browser:
file:///home/jovyan/.local/share/jupyter/runtime/jpserver-7-open.html
Or copy and paste one of these URLs:
http://98abeeadacd0f:8888/lab?token=6959babcf67d78f2eddc95dbcaa863365c93fe033e64e557
or http://127.0.0.1:8888/lab?token=6959babcf67d78f2eddc95dbcaa863365c93fe033e64e557
Joans-MacBook-Air:jupyter-work joanribot$ docker exec -it jupyter bash
(base) jovyan@98abeeadacd0f:~$ ls /home/jovyan/work
bind.ipynb
Joans-MacBook-Air:jupyter-work joanribot$ mv bind.ipynb /work/
Joans-MacBook-Air:jupyter-work joanribot$ exit
Joans-MacBook-Air:jupyter-work joanribot$ docker stop jupyter
jupyter
Joans-MacBook-Air:jupyter-work joanribot$ docker rm jupyter
jupyter
Joans-MacBook-Air:jupyter-work joanribot$
```